

THOMAS ISTVAN

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SUMMARY

Robotics & Controls Engineer with hands-on experience across mechanical design, embedded control software, and multi-sensor state estimation for autonomous ground, aerial, and marine robots. MEng candidate in Mechanical Engineering at Cornell (Summa Cum Laude BS); builds ROS 2 control architectures, EKF sensor-fusion pipelines, and precision mechanisms from CAD through fabrication.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Aug 2026 – May 2027

Master of Engineering in Mechanical Engineering

MEC Fellowship (\$30,000) and Academic Tuition Scholarship (\$6,000)

University of Hawai'i at Mānoa, College of Engineering, Honolulu, HI

Aug 2023 – May 2026

Bachelor of Science in Mechanical Engineering, **Summa Cum Laude, GPA: 3.98**

Manoa Academic Scholarship (\$12,000) and ASUH Research Scholarship (\$1,000)

Relevant Coursework: Robot Perception, Legged Robots, Model-Based Estimation, Autonomous Mobile Robots, Swarm Robotics, Maritime Robotics, Soft Robotics, Manipulator Robotics

SKILLS

Robotics & Controls: ROS2, PID control, forward/inverse kinematics, Jacobian-based force control, Extended Kalman Filter (EKF) sensor fusion, state estimation, embedded systems, PWM servo actuation

Software: Python, C++, MATLAB, Linux, PyTorch, MuJoCo, Gazebo, RViz2, Arduino IDE

Mechanical & Hardware: SolidWorks, Onshape, COMSOL, CNC, mill, lathe, bandsaw, drill press, FDM & resin 3D printing, Arduino, Raspberry Pi, Teensy, I2C/UART/SPI/USB

EXPERIENCE

Software Engineer, ASML, Ithaca, NY

Sept 2026 – Present

- Re-architect a legacy LabVIEW control system into a modular ROS 2 stack for a reticle-handling robotic arm by reverse-engineering the existing control logic and rebuilding it as custom C++ nodes.
- Validate control logic before hardware deployment by pairing the ROS 2 control architecture with a physics-based simulation test harness in Gazebo, reducing the risk of on-hardware failures.
- Enable closed-loop velocity control of a multi-axis robotic arm by deriving forward/inverse kinematic models and Jacobian-based force mapping that translate end-effector targets into real-time joint-space commands.

Mobile Robotics Research Assistant, RIP Lab, Honolulu, HI

May 2024 – July 2026

- Achieve stable vertical-surface operation for a 50-lb climbing robot by optimizing magnetic-adhesion geometry through MATLAB and COMSOL Multiphysics simulation prior to fabrication.
- Cut prototype iteration time by rapid-prototyping sensor housings and caster wheels in Onshape and producing them in-house via FDM and resin 3D printing.
- Provide real-time orientation and state feedback during testing by building a C++ ROS 2 pipeline that fuses dual 9-DOF IMU streams through an Extended Kalman Filter and renders the fused state live in RViz2.
- Protect the robot's electronics from thermal failure during extended runtime by building a Python monitoring tool that tracks battery-enclosure temperature within 5% of a reference sensor.

Lead Software Engineer, Team Kanaloa, Honolulu, HI

Aug 2025 – May 2026

- Lead a student team building an autonomous surface vessel for littoral zone research, modeling the vehicle in URDF/XML and validating the C++ PID control and autonomy stack in Gazebo before on-water testing.
- Achieve centimeter-level real-time localization accuracy for the autonomous vessel by implementing an EKF sensor-fusion node that fuses GPS, IMU, and wheel-encoder measurements with RTCM correction.
- Bring the vessel's GPS, IMU, motor, and onboard electronics online for the autonomy stack by writing low-level ROS2 drivers and hardware interfaces over I2C, UART, SPI, and USB.

Soft Robotics Research Assistant, SAIL Lab, Honolulu, HI

May 2024 – May 2025

- Enable repeatable soft-robotic fin motion for lab testing by writing embedded C++ control software on Arduino to automate PWM-driven pneumatic servo actuation.
- Achieve >90% accuracy against 3-axis magnetometer ground truth by building a Python tool that analytically computes the 3D magnetic field of an N52 magnet at arbitrary points.
- Iterate pectoral fin prototypes for swimming performance by designing them in Onshape and fabricating FDM-printed molds cast in Dragon Skin silicone rubber.

LEADERSHIP EXPERIENCE

Graduate Teaching Assistant, Mechanical Engineering Department, Cornell University

Aug 2026 – Present

- Translate control-theory coursework into hands-on lab exercises by teaching open- and closed-loop feedback control systems to 21 undergraduates across weekly labs and office hours.

ACTIVITIES & INTERESTS

Pi Tau Sigma Honor Society, American Society of Mechanical Engineers, Institute of Electrical and Electronics Engineers | Pickleball, basketball, hiking, watching NBA and CFB